Team No 67 - (LLOYD Problem B)

1. Current approach and changes since M1

Our approach is the same as in M1, and it is now fully built and working end to end.

- **A simulator.** It generates our training data. It creates 30,000 realistic made up homes by picking features including region, house type, age, size, walls, loft, windows and heating.
- **Machine-learning models (LightGBM).** We used these to estimate a home's energy use when the customer gives incomplete or uncertain answers. They also give a likely range, not just a single number.
- The scores are calculated to map accurately to its physical meaning.
- **A retrofit planner.** It first removes improvements that don't suit the home, then tests every combination of the remaining ones together, because improvements affect each other. It then picks the best combination for the customer's goal and budget.
- **The LLM:** It only explains results and answers questions based on data processed by the engine. A checker blocks any answer containing a number the engine didn't produce.

The whole pipeline runs with one command (make all): it creates the data, trains the models and writes the evaluation report.

Changes since M1:

- **Dashboard:** we used Streamlit instead of React, because it let us build faster. It has five screens: home overview, retrofit planner, plan results, what-if (sliders and chat), and assumptions.
- **LLM provider:** We are using an api key from Gemini. If Gemini is unavailable, the system falls back to fixed sentence templates, so the demo always works.

2. Experiments run so far and their results

Data: 30,000 made up (synthetic) homes. They are split by customer so the same customer never appears in both training and testing: about 70% for training, 15% for validation and 15% for testing. A separate extra hard test set contains only homes from a region and a building era the model never saw in training (Wales, and homes built 1900-1929).

- **Predicting energy use.** Average error while testing has been about 21% for electricity and 16% for gas. A simple guess (the average for that house type and age) is off by about 53% and 51%. Accuracy holds on the extra hard test set (21% and 17%).
- **Uncertainty ranges.** We aimed for ranges that contain the true value 80% of the time. They do so 78-80% of the time, so the ranges are reliable.
- **What helps the model.** Removing the physics based inputs makes it worse (electricity error rises from 709 to 762 kWh). The ML model does about as well as the simulator on its own (704 kWh) since our test answers come from the same simulator. The model's real value is handling unsure answers and giving trustworthy ranges.
- **Which improvements suit a home.** The suitability checks are 95-100% correct on both measures (precision and recall). Always returning no finds none of the suitable homes.
- **Finding homes with the most room to improve.** Our ranking agrees closely with the reference answer (0.95 out of 1). Ranking by EPC band alone barely agrees (0.10).
- **Planner.** Checking every combination gives on average 3.6% bigger bill savings than a quicker one at a time method, and that method makes a worse choice for about half of

homes. A full plan takes about 7 ms to build, and changing goal or budget takes about 21 ms.
- **Robustness.** When we change the simulator's settings by up to 20%, the gas error grows from 1,628 to 2,119 kWh. So our results depend on getting those settings right.
- **Property value.** This currently uses practice sales data we generated ourselves. Price error is 15% against 27% for a simple average. The ranges contain the true price 73% of the time, below our 80% target.

3. Progress on testing

- **Automated testing:** We ran 59 automated tests, all of which passed. These covered energy balance for solar, batteries, grid imports and exports, non negative energy values, insulation and heat pump effects, score behaviour including the problem statement's example score of 38, and handling of missing or conflicting data, budgets, tenants and listed buildings. We also tested the number checker and web API.

- **Edge cases:** Missing information gets a specific question instead of a guess. Conflicting data, such as a heat pump alongside 12,000 kWh of gas use, asks for clarification. Unsuitable improvements are explained without claiming savings, and increased electricity use after a heat pump is clearly flagged.

- **AI evaluation:** We tested **57 questions** covering explanations, what-ifs, budgets, missing or conflicting data, and unsuitable improvements.

    ➔ **Number checker:** 100% pass rate

    ➔ **Groundedness:** 73.7% (target 90%)

    ➔ **Factual accuracy:** 78.9% (target 90%)

    ➔ **Unsupported claims:** 5.4% of sentences (target under 5%)

- Most errors appeared in what-if questions, especially price changes, tenants, solar size, and heat pumps. The AI sometimes added questions or described changes that were not provided. Plain explanations, unsuitable improvement cases, and missing or conflicting data responses were correct.

4. Plan for the remaining time

1. Increase the LLM accuracy. Give the AI the details of each what-if change directly, and tell it to ask only the questions our system flags. Rerun and remark the questions in our test file, aiming for the target given in the problem statement.
2. Better explanations. Show where a home loses its heat (walls, windows, draughts) and which inputs drive the AI model's estimate.
3. More testing needs to be done on diverse values.
4. Improve the user experience and make the plots and statistics more understandable for the customer.